

Large Language Models and intraday financial forecasting



Evidence from a
Real-Time Out-of-
Sample Experiment

MARTINA GAMBINA

LUGLIO 2026



UNIVERSITÀ
DI PARMA

Research Question



I Large Language Models possono generare segnali predittivi utili per il forecasting intraday?



Le previsioni degli LLM sono economicamente stabili?



I risultati resistono a un framework real-time e out-of-sample?

NO LOOK- AHEAD BIAS: ALL PREDICTIONS ARE MADE BEFORE MARKET OPEN USING ONLY PUBLICLY AVAILABLE INFORMATION.

Real-time forecasting environment



1. Prompt

Pre-market information only



2. LLM Prediction

Long/Short,
Market timing,
Price target



3. Market Open

U.S. market session begins



4. Market Close

Intraday returns observed



5. Evaluation

Portfolio construction and metrics

Literature Background

The intersection of Market Efficiency and Artificial Intelligence.

1 Market Efficiency & Return Predictability

Efficient Market Hypothesis (EMH): Public information is rapidly incorporated into asset prices.

Predictability limits: Consistent abnormal returns are difficult to achieve.

Delayed Reactions: Predictability, when present, is weak, short-lived, and often caused by cognitive biases or complex information processing.



2 Information Processing & Textual Analysis

News impact: Information arrives primarily through financial news and public corporate disclosures.

Methodological shift: Transition from simple dictionary-based sentiment to advanced Machine Learning.

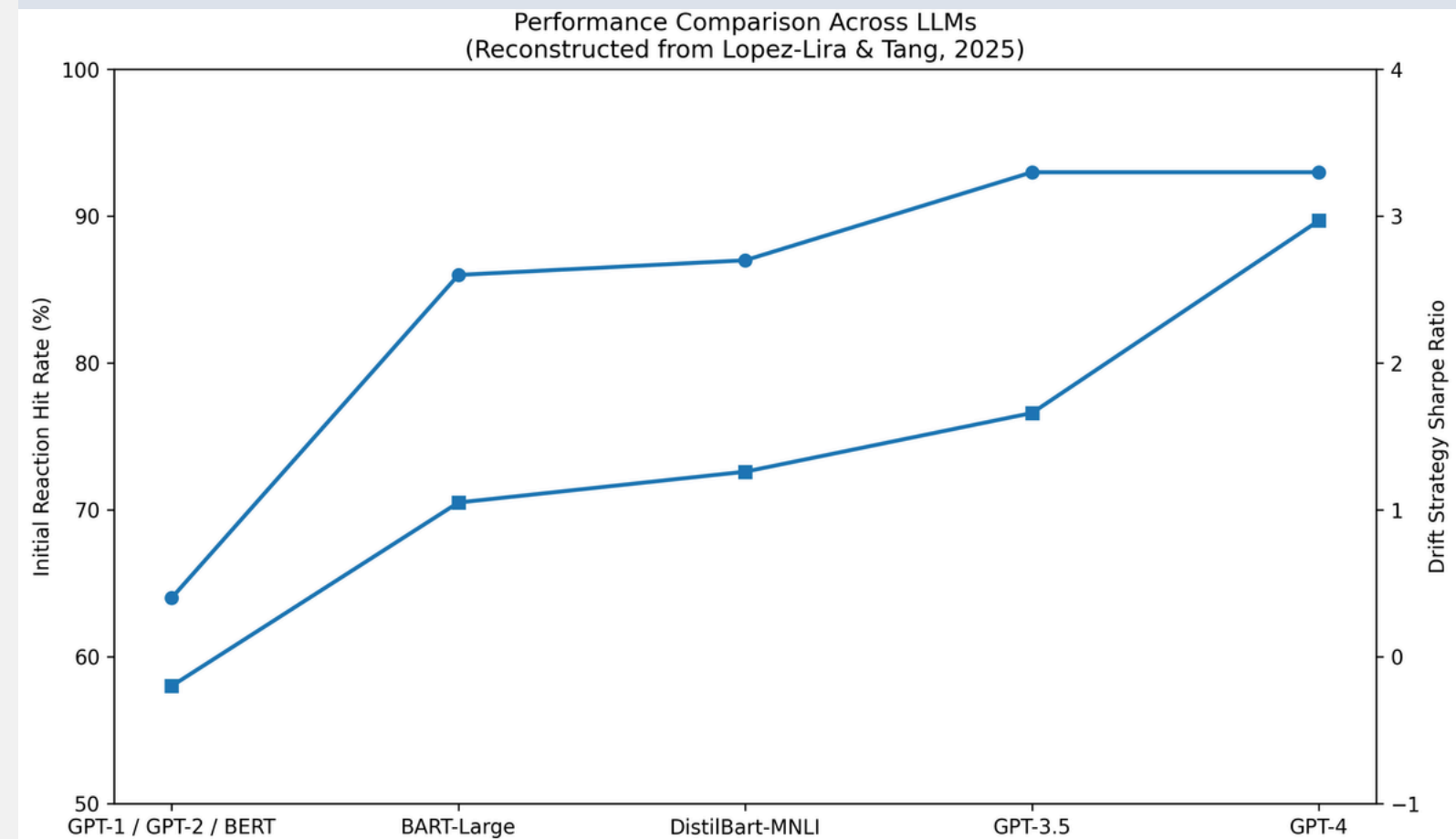
Objective: Capturing market sentiment and underlying investor attention automatically.



3 LLMs & Financial Forecasting

Transformer architecture: Allows for deep semantic understanding of complex financial narratives.

Evidence (Lopez-Lira & Tang, 2025): Advanced LLMs can predict market reactions from news headlines. Predictive performance increases alongside model sophistication.



SOURCE: LOPEZ-LIRA & TANG (2025), TABLE 5

Empirical Framework & Research Design

Strict real-time protocol for intraday forecasting via LLMs: The strict real-time protocol ensures the total elimination of look-ahead bias. Every prediction is generated on public information available before asset prices are realized.



Portfolio Strategy & Evaluation Framework

Weighting schemes, predictive and risk-adjusted performance metrics.

Portfolio Construction

- 1

Equally Weighted (EW)

- Equal Exposure across positions
 - Reduces concentration effects
- 2

Signal Weighted (SW)

- Weights based on explicit LLM ranking (1-3)
 - Stronger signals receive larger allocations
- 3

Value Weighted (VW)

- Weights based on market capitalization
 - Closer to institutional implementation

Performance Evaluation

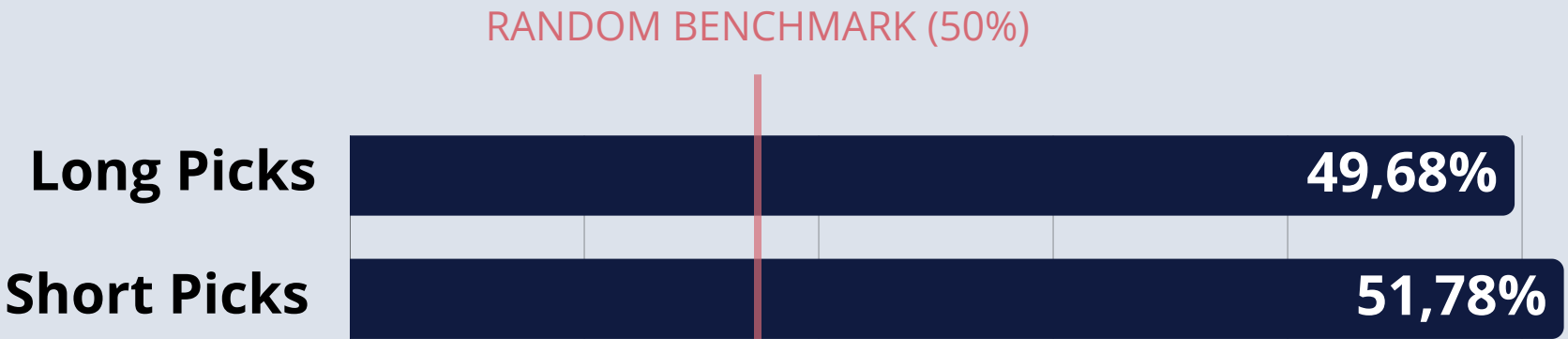
	Intraday Asset Returns	Open-to-Close
	Hit Ratio	Directional Accuracy vs. 50% Random Benchmark
	Sharpe Ratio	Risk-Adjusted returns (Daily frequency)
	Jensen's Alpha	Abnormal performance via CAPM intraday
	4-Factor Alpha	Fama-French + Momentum risk control

RISK-ADJUSTED & PREDICTIVE METRICS

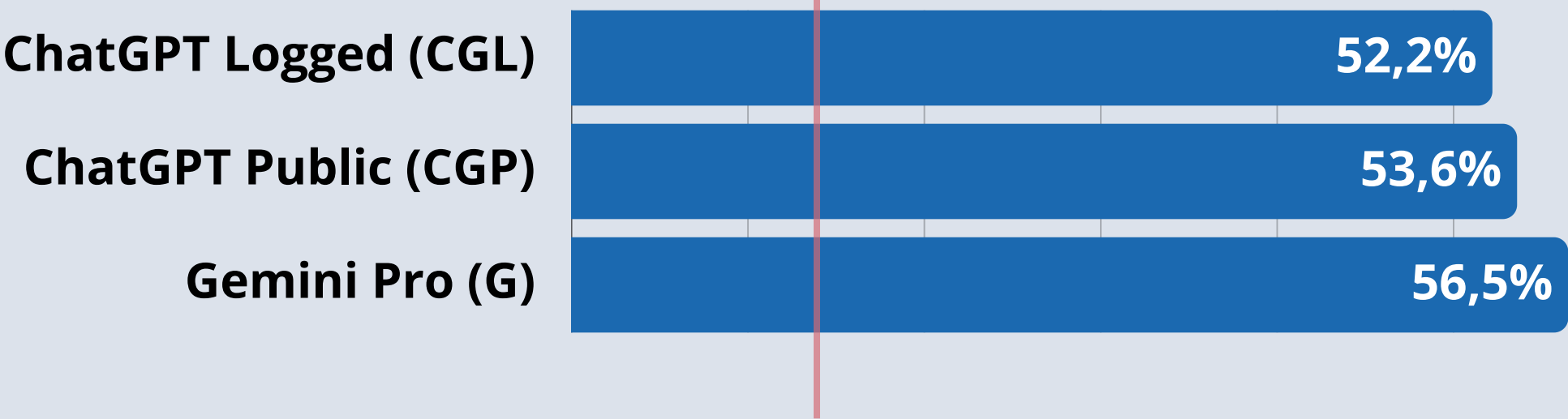
Sharpe Ratio	$SR = \frac{E(R_p - R_f)}{\sigma_p}$
CAPM	$R_{p,t} - R_{f,t} = \alpha_p + \beta (MktRF_t^{intraday}) + \varepsilon_t$
Intraday MKT excess return	$MktRF_t^{intraday} = R_t^{MKT} - RF_t$
Fama-French-Carhart	$R_{p,t} - R_{f,t} = \alpha_p + \beta_1 MKTRF_t + \beta_2 SMB_t + \beta_3 HML_t + \beta_4 MOM_t + \varepsilon_t$
Magnitude Forecasting	$R_{i,t}^{real} = \alpha + \beta R_{i,t}^{pred} + \varepsilon_{i,t}$

Directional Accuracy & Market Timing

HIT RATIO vs. RANDOM WALK (50%)



MARKET TIMING (S&P 500 INDEX)



INFERENTIAL ANALYSIS

Statistical significance via One-Sample T-test and Binomial Tests:

p-value Long Picks: 0.873

p-value Short Picks: 0.377

p-value Market timing (CGL): 0.810

p-value Market timing (CGP): 0.630

p-value Market timing (G): 0.336

Statistically not significant

All predictive outputs are statistically equivalent to a random walk ($p > 0.05$).

Note: Rigorous out-of-sample analysis conducted in a real-time information environment

Cumulative Portfolio Performance

Model Heterogeneity

Marked divergence in performance: **ChatGPT Logged** shows systematically weak or negative returns throughout the sample period.

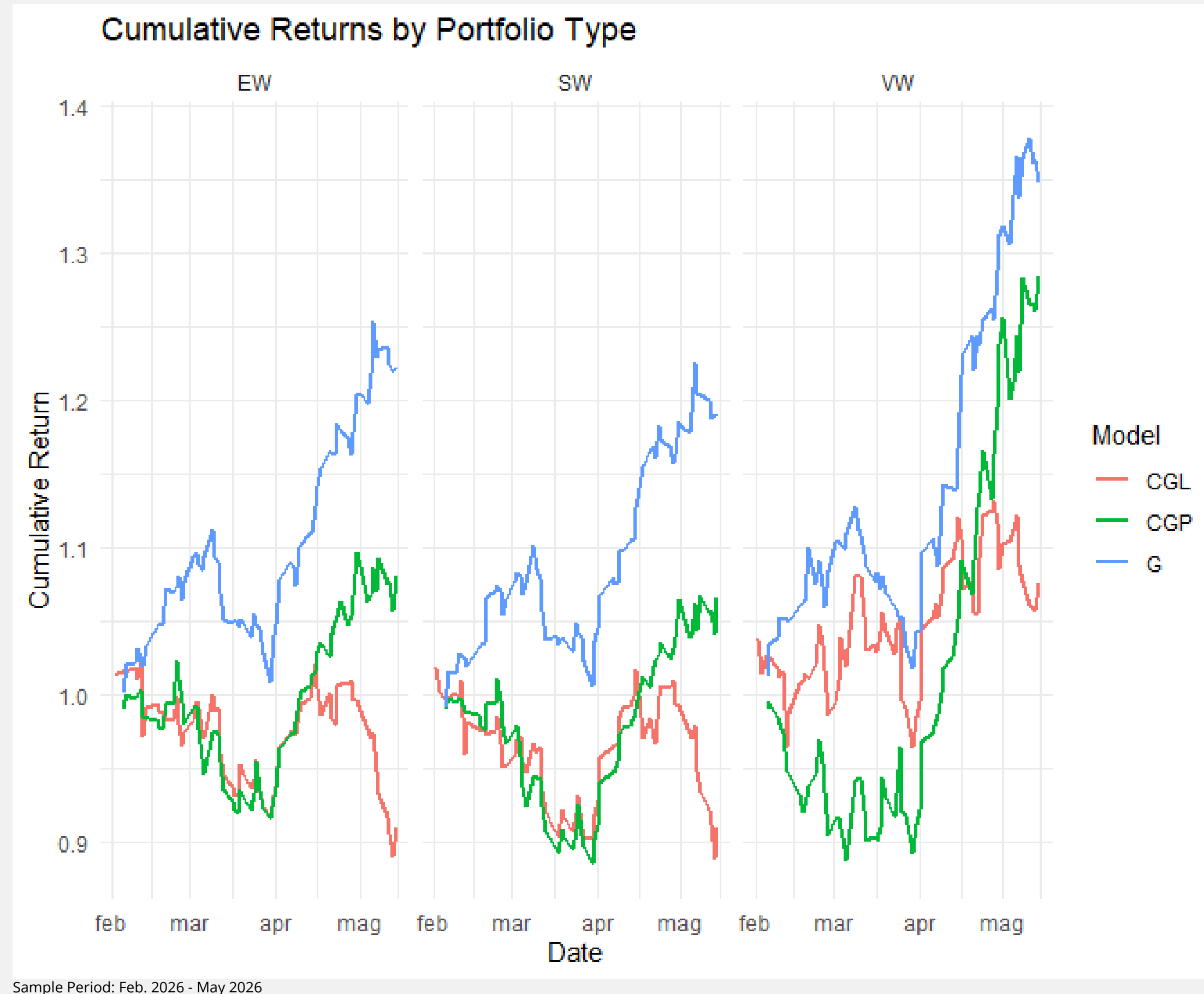
The Dominant Performer

The Value-Weighted (VW) portfolio based on **Gemini Pro (G)** signals outperforms all other configurations, reaching a **+34.7% Cumulative Return**.

Weighting Impact

Value-Weighted (VW) schemes generally yield higher returns than EW or SW counterparts, suggesting institutional relevance.

Interpretative Caution: Performance is likely driven by **systematic factor** exposure (High-Beta/Momentum) rather than idiosyncratic stock selection skill.



Daily Return Distributions & Signal Stability

Return Density Mapping

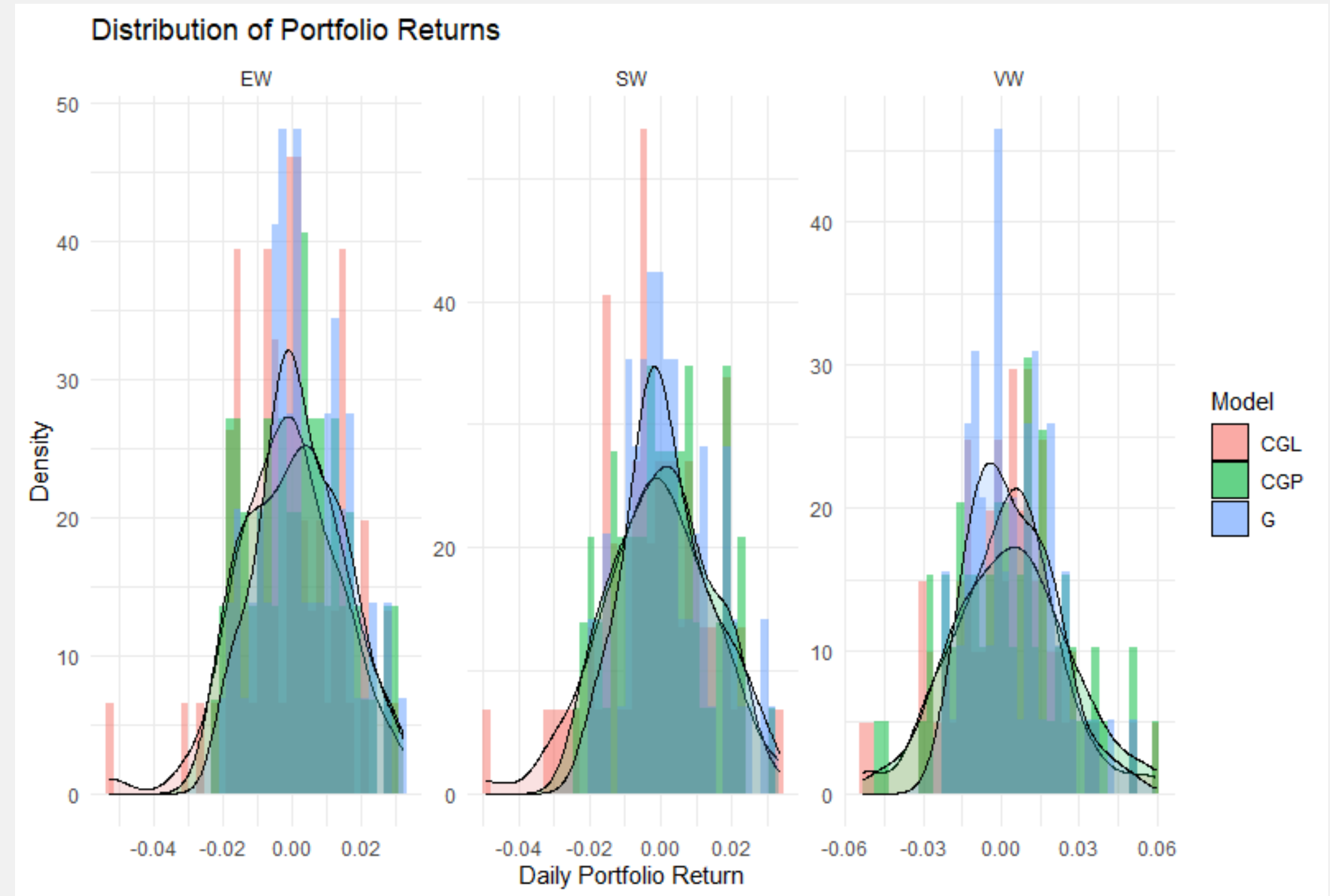
Cross-selection distribution analysis confirms distinct structural behaviors across LLM architectures rather than isolated outlier anomalies.

The Value-Weighted (VW) Shift

Gemini Pro (G) exhibits a rightward distributional shift and higher peak density under VW, validating its ability to capture systematic large-cap trends.

Dispersion & Tail Risk

ChatGPT configurations display higher return variance and left-skewness, reflecting a less stable inferential framework for daily rebalancing.

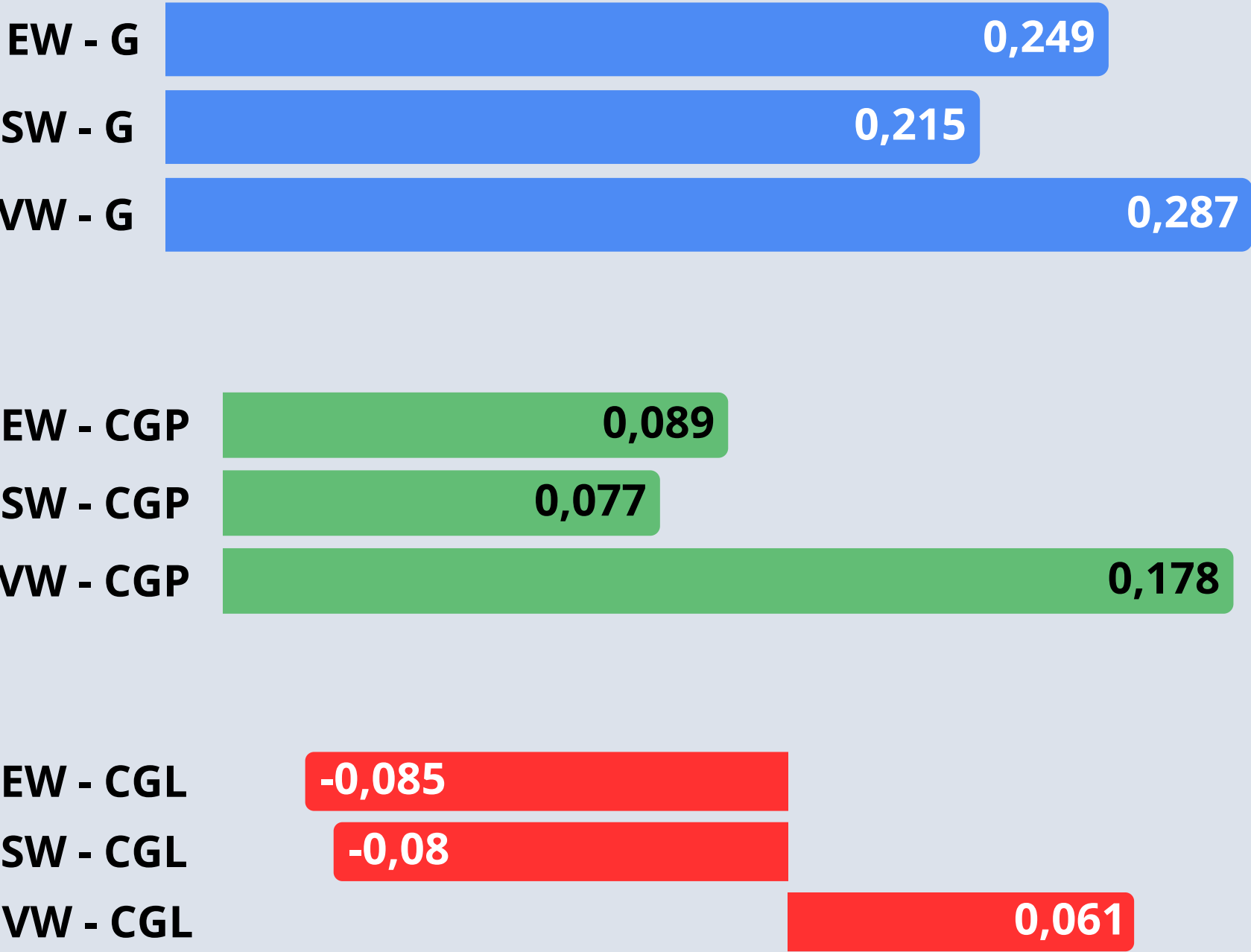


Note: Empirical distribution of daily portfolio returns. Real-time out-of-sample data (Feb-May 2026)

Risk-Adjusted Performance: Sharpe Ratio

Daily Sharpe Ratio breakdown across the three portfolio weighting schemes.

Daily Sharpe Ratio by Configuration



Superior Efficiency

Gemini Pro (G) consistently dominates all other configurations in terms of risk-adjusted returns, particularly in the **Value - Weighted (VW)** scheme.

Risk-Return Trade-off

The increase in absolute returns observed in VW portfolios is accompanied by higher volatility, yet the Sharpe Ratio confirms a **net gain in efficiency** for the Gemini model.

Comparative Modesty

Despite being positive, these ratios remain **modest** compared to established quantitative strategies, suggesting that LLM signals are not yet robust enough for stand-alone implementation.

Note: Sharpe ratios are daily and non-annualized (R_f=0). Ratios remain modest compared to institutional quantitative benchmarks

Magnitude Forecasting: Quantitative Estimates

ChatGPT: The “Inverse Bias” Paradox

Regression analysis reveals a counterintuitive dynamic: more aggressive numerical estimates from ChatGPT correspond to **worse** actual market performance. The model systematically misinterprets the magnitude of trends.

Gemini Pro: Statistical Noise

Unlike ChatGPT, Gemini Pro does not exhibit this systematic negative bias; its quantitative point estimates behave essentially as **statistical noise**, showing no significant correlation with realized returns.

	CGL	CGL	CGP	CGP	G	G
	Intercetta	Predicted Return	Intercetta	Predicted Return	Intercetta	Predicted Return
Estimate	0.00123	-0.00165**	0.00470**	-0.00217**	0.00301	0.00006
Std. Error	0.00208	0.00072	0.00205	0.00098	0.00186	0.00055
t-stat	0.590	-2.290	2.290	-2.220	1.620	0.110
p-value	0.556	0.023	0.023	0.028	0.107	0.912

Note: R² < 1% for all models. Analysis of quantitative numerical estimates extracted directly from LLM prompt

Conclusion: LLMs are currently incapable of generating reliable numerical point estimates for asset returns

Asset Pricing: Alpha & Risk Factors

Regression Coefficient - Gemini PRO VW Portfolio

Variable	CAPM Model	FFC 4-Factor Model
Alpha (Intercept)	+0.00343**	0.00236
MKT_RF_intraday	1,280***	-
MKT_RF	-	0.613**
SMB	-	-0.245
HML	-	-0.232
MOM	-	-0.131

The Alpha Decay

Under the CAPM framework, Gemini Pro exhibits a positive and significant Alpha (**+0,34% daily**). However, this statistical significance completely vanishes (ns) in the multifactor model.

This demonstrates that abnormal returns are fully absorbed by systematic risk factors rather than stemming from an intrinsic stock-picking ability.

Momentum Divergence

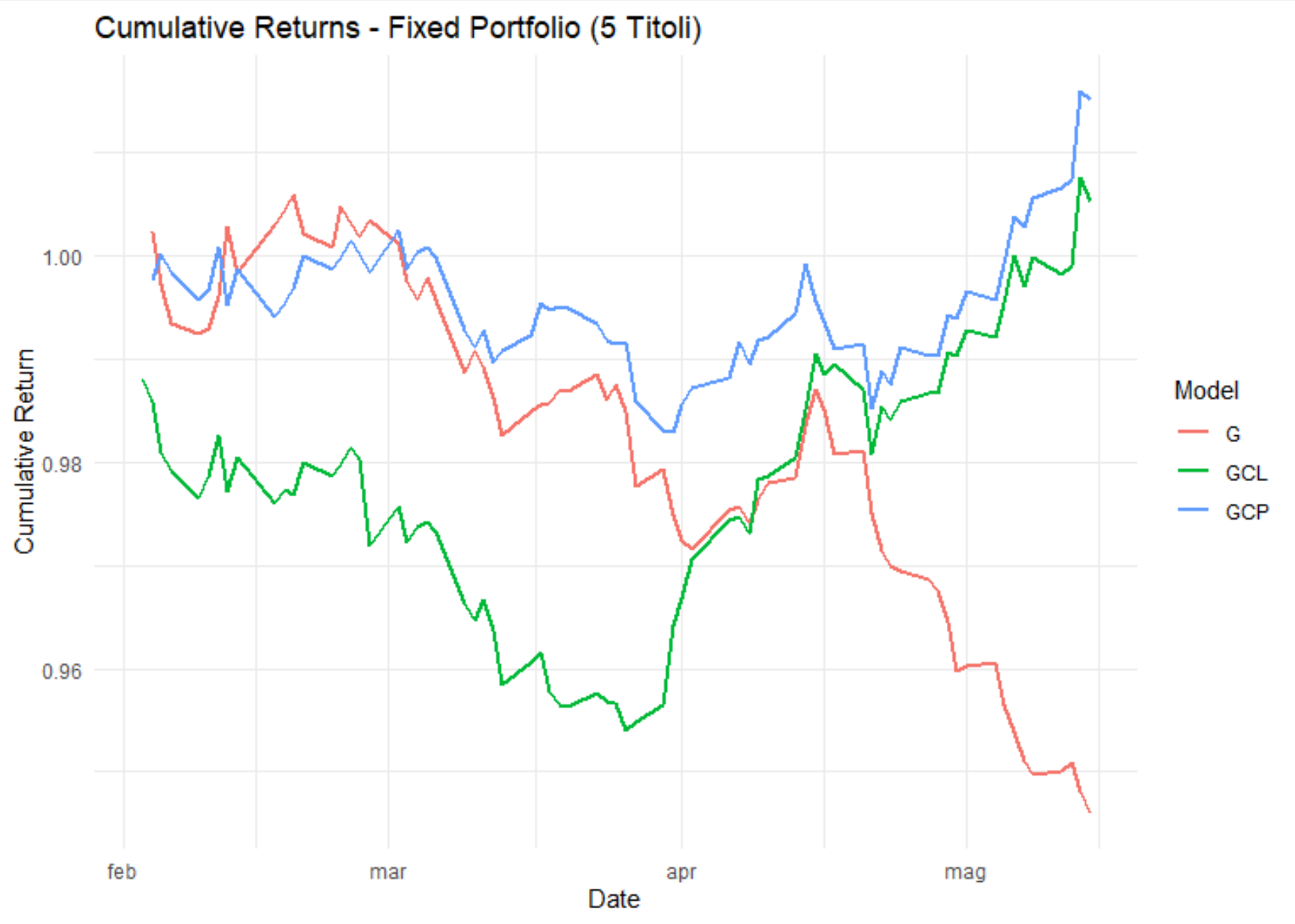
While the ChatGPT model shows an extremely strong and highly significant loading on the Momentum factor (**0,908*****), Gemini Pro presents a neutral and non-significant exposure (-0,131), highlighting structurally different predictive logics.

$$R_{p,t} - R_{f,t} = \alpha_p + \beta_1 MKTRF_t + \beta_2 SMB_t + \beta_3 HML_t + \beta_4 MOM_t + \varepsilon_t$$

Methodological Robustness Checks

These findings confirm the stability of LLM signals against sample bias and statistical anomalies.

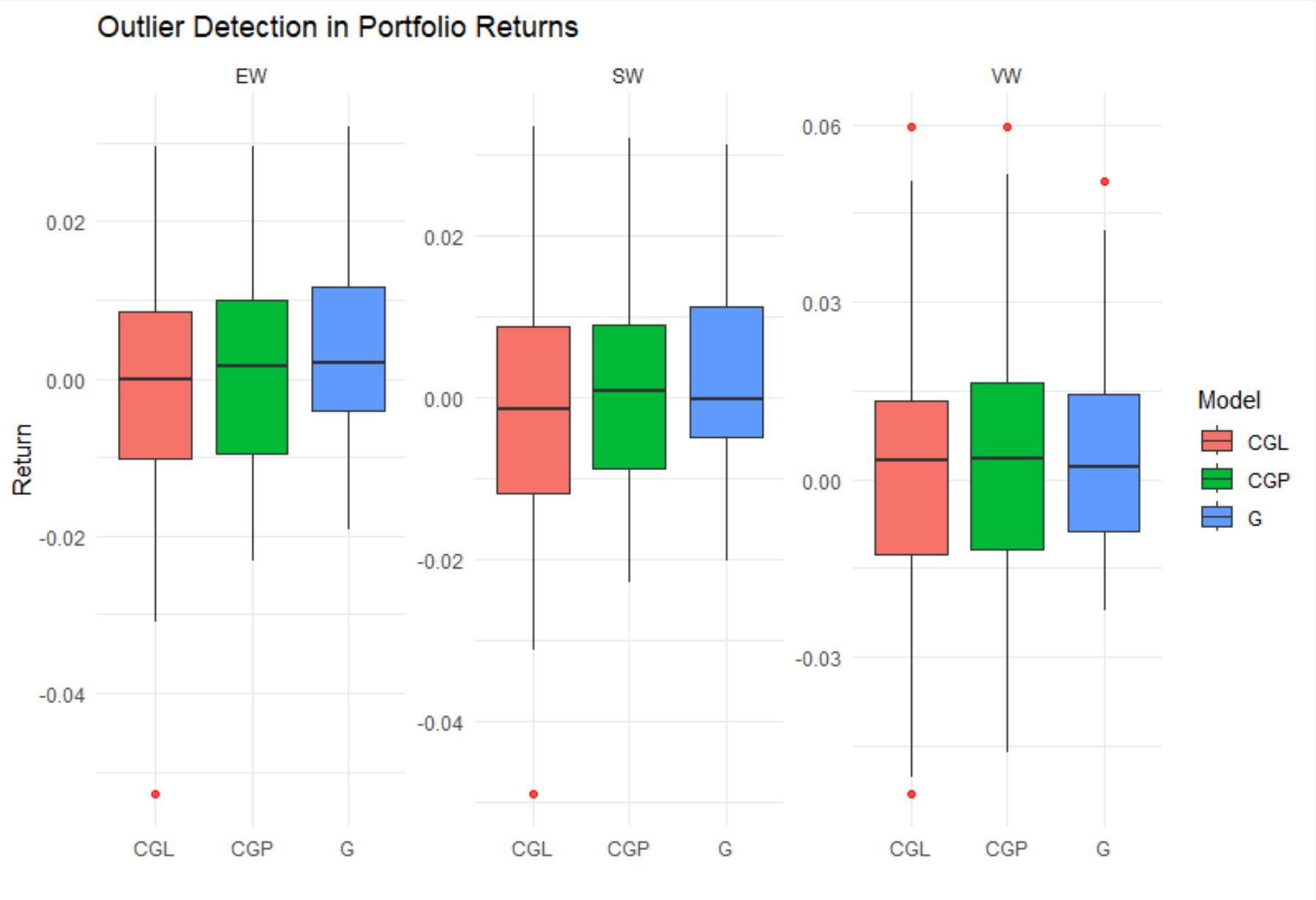
Fixed Portfolio Analysis



Testing performance on a **fixed investment universe** (NVIDIA, Apple, Microsoft, Amazon, Alphabet A.) to neutralize selection biases. This confirms Alpha is not driven by dynamic sample changes.

Result: Performance ranking (G > CGP > CGL) remains stable.

Outlier Management (IQR)



Application of the **Interquartile Range (IQR)** method to filter extreme return anomalies. The strategy's success is structural, not accidental. Removing severe market anomalies does not alter the positive Sharpe ratios or the core findings.

Result: Cumulative returns and Sharpe ratios remain structurally robust.

Conclusion & Future Outlook

The illusion of True Alpha

While Gemini Pro appears to generate abnormal returns under a simple CAPM framework, this outperformance completely vanishes in multi-factor models. LLM returns are absorbed by systematic risk factors, capturing market premiums rather than displaying idiosyncratic stock-picking genius.

Quantitative point-estimate failure

LLMs remain highly unreliable for predicting exact price magnitudes. ChatGPT models exhibit a counterintuitive and significant "**Inverse Bias Paradox**", while Gemini Pro behaves as statistical noise, highlighting severe limits in text-to-numerical conversion.

Architectural & Style Divergence

Distinct behavioral footprints emerge across models. ChatGPT strategies are heavily driven by the market **Momentum factor (0,908***)**, whereas Gemini Pro maintains neutral factor exposure while delivering superior risk-adjusted stability (Max Sharpe of 0,287).

Methodological Robustness

Robustness checks confirm that these findings reflect structural model behavior rather than artifacts. Results remain consistent when restricted to a static universe and after filtering extreme operational outliers using IQR.

Future Research Direction: Financial integration of Large Language Models should pivot away from zero-shot numerical return forecasting. Instead, future frameworks should leverage LLMs exclusively for **qualitative sentiment extraction and directional signals**, leaving magnitude calibration and portfolio optimization to traditional econometric frameworks.